Ben Pilger

Systems & Security Engineer | IT Trainer

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Core Competencies

Languages: Rust, C++, C, x86-64 Assembly, Python

Domains: Zero-Trust Architecture, PAM, IAM, Cryptography, PKI, Systems Security, Reverse Engineering, Kernel Development, Debugging & OS Internals (Windows/Linux)

Frameworks & Networking: Tokio, Actix, gRPC, REST, TCP/IP, TLS, AWS, Cloudflare

Tooling: x64dbg, IDA Pro, Ghidra, WinDbg, WDK, CMake, Visual Studio, Docker, CI/CD

Professional Experience

Freelance Engineering & IT Training
Stuttgart, Germany

Independent Consultant & Technical Trainer
07/2026 – Ongoing

- **Systems Engineering Consultancy:** Provide specialized research, architectural auditing, and low-level development in Rust, C++, and Assembly, focusing on high-concurrency tooling and security hardening.
- **Trainer:** Available for corporate seminars covering low-level kernel programming (Windows/Linux), networking (TCP/IP), and complex Rust FFI integrations.

AnyDesk Software GmbH
Stuttgart, Germany

Backend Engineer – Solutions Engineering
04/2025 – 09/2026

- **Decentralized PAM Architecture:** Pioneered a novel Privileged Access Management (PAM) infrastructure by conceptualising a peer-to-peer trust model that completely eliminated architectural Single Points of Failure (SPOFs).
- **Cryptographic Protocol Engineering:** Designed and implemented the cryptographic backbone for the decentralized PAM ecosystem; developed a custom asymmetric key-based configuration protocol and a decentralized PKI-service to enforce end-to-end integrity and tamper-proof state synchronization.
- **System Modernization & Stability:** Contributed to the migration of legacy C++, Erlang, and TypeScript services to a high-performance Rust-based microservice architecture, optimizing for high-concurrency environments (>200K RPS) and strict sub-100ms latency guarantees.
- **Enterprise Product Leadership:** Acted as a technical lead for complex enterprise requirements, managing the full lifecycle from requirement engineering and protocol specification to the delivery of scalable IAM and fraud mitigation services.

Freelance R&D / Self-Employed
Stuttgart, Germany

Systems & Security Engineer
11/2024 – 03/2025

- **Low-Level Kernel Engineering:** Developed system-level drivers and modules interacting with the Windows kernel using Rust and C++. Performed deep-dive binary analysis and low-level debugging of core operating system architectures.
- **Enterprise Platforms:** Architected a highly scalable sales, user, and license management infrastructure with integrated international payment gateways and automated client software lifecycle deployment.

Sparx.Foundation
Global

Research
09/2024 – Ongoing

- **Threat Emulation Framework:** Architected and engineered an advanced network resilience and threat emulation platform (”Actrax”) in Rust and C++ to evaluate detection engineering capabilities and endpoint defense mechanisms.

Independent Security Researcher
Remote

Offensive Tooling & Compatibility Engineering 03/2023 – 08/2024

- **Mass Deployment & Compatibility (05/2024 - 08/2024):** Deployed and maintained custom software for a global user base, ensuring continuous runtime stability across hundreds of volatile Windows OS builds and custom hardware environments.
- **Mission-Critical Infrastructure Utilities (12/2023 - 04/2024):** Engineered a custom suite of infrastructure health utilities and configuration verification tools designed to validate environment states and ensure maximum availability in high-security, sensitive deployment environments.
- **Reverse Engineering & Integrity Bypasses (03/2023 - 11/2023):** Analyzed complex security mechanisms and runtime protections using IDA Pro and WinDbg to evaluate functional software integrity and binary execution flows.

Technical Training

IT Trainer & Course Instructor
Stuttgart, Germany

Specialized Technical Curriculum Delivery 07/2026 – Ongoing

- **Networking & Protocol Seminars:** Deliver comprehensive training tracks covering **Foundational Networking & TCP/IP** for beginners, extending up to deep-dive **Advanced TCP/IP Stack Analysis** and specialized enterprise network infrastructure seminars.
- **Windows Operating System Internals:** Conduct deep-dive courses spanning **Windows System Programming**, core **Windows Internals** architectures, and premium developer tracks covering **Windows Kernel Programming** and driver development.
- **Linux Engineering & Architecture:** Teach comprehensive modules on **Linux System Programming** and **Linux Internals** (including Enterprise Linux environments like RHEL), scaling into specialized **Linux Kernel Programming** and kernel-mode module development.
- **Cybersecurity & Offensive Seminars:** Lead high-impact security workshops covering **Network Security**, **Web Application Security**, and high-ticket technical tracks focusing on **Malware Analysis**, binary executable exploration, and advanced **Reverse Engineering**.

Research & Open Source Projects

Systems & Security Research Platform
arteii.net

Technical Author 12/2023 – Ongoing

- **Windows Kernel Security Research:** Published low-level technical documentation and analysis covering Ring-0 driver architecture, exception handling routines, and hardware-assisted mitigations (IOMMU, Virtualization-Based Security).
- **Low-Level Systems Analysis:** Authored mathematical and technical references targeting x86-64 assembly conventions and machine-level instruction sets.

GitHub Ecosystem *Core Contributor & Maintainer*

- **lychee:** Optimized engine performance and connection handling for the widely used asynchronous link-checking engine written in Rust.
- **zenity:** Maintained and scaled a CLI UI framework in Rust, backed by JetBrains OSS sponsoring.
- **RegistryHelper** / **RegistryGuard:** Authored a C++ helper tool designed for Windows registry configuration access and real-time modification tracking via thread-based background callbacks.
- **limitr:** Built a high-throughput rate-limiting and resource management utility for concurrent systems.
- **web_client_lib:** Implemented a modular network library for streamlined, secure HTTP communications.